

Complete Merkle tree (height 4) with stored proof-bearing nodes highlighted

- default empty node
- real but not stored
- occupied leaf
- stored node C1
- stored node C2

Example settings
height = 4
total leaves = 16
occupied leaves = 3
bottom-layer slots = [0, 11, 15]
sparsity = 81.2%

Coloring result
colors = 2
stored proof nodes = 4
count loads = [2, 2]
weight loads = [3, 2]
count gap = 0
weight gap = 1

Sample batch-PIR views
leaf heap=16 slot=0: C1->3, C2->dummy
leaf heap=27 slot=11: C1->2, C2->7
leaf heap=31 slot=15: C1->2, C2->6

